



Stove Sensor

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Introduction



- Over 55 million people worldwide suffered from dementia in 2020, and is estimated to reach 139 million in 2050 [1]
- A large portion of these patients lack proper access to healthcare, leaving them to rely on family or by themselves
- There is an increased risk to kitchen hazards from stoves being accidentally left on



Solution



- Multiple sensors that each detect signs that the stove may have been accidentally left on
- A consistent alarm system that reminds users to turn off the stove



Methods



Sensors

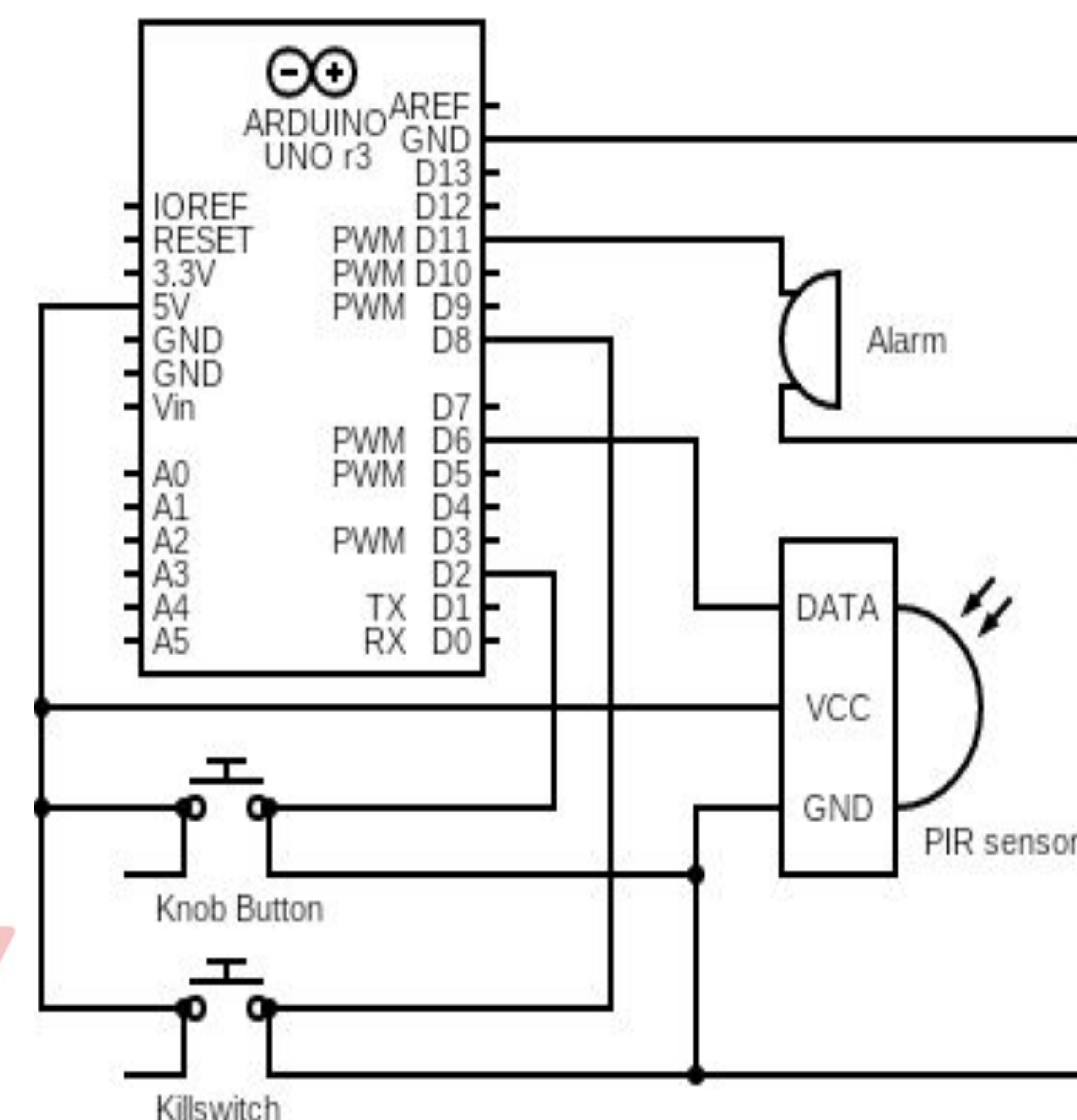
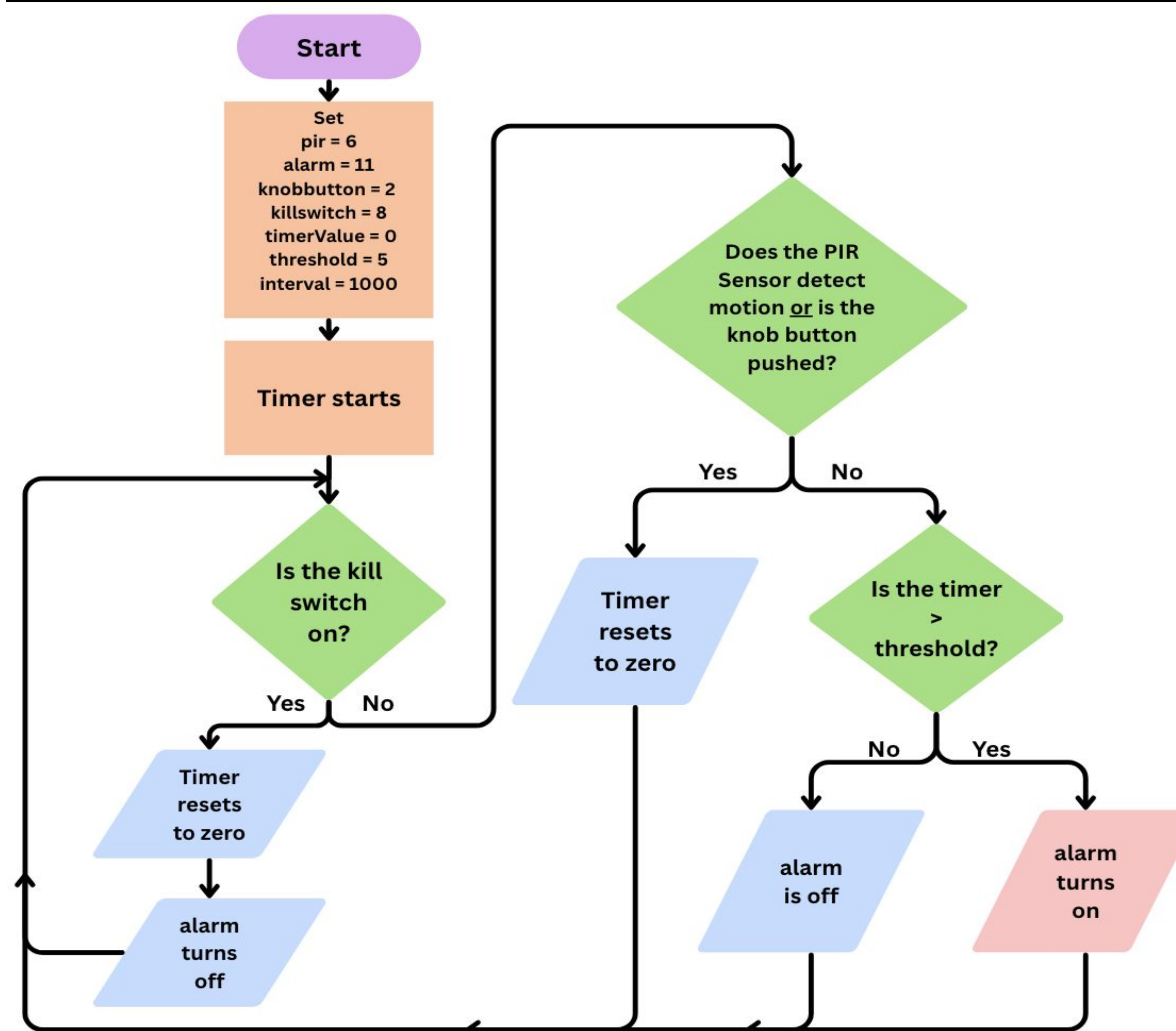
- An attachment to the stove knob releases a button when the stove is turned on
- A PIR sensor detects motion in front and above the stove

Outputs

- When a certain amount of time has passed with the stove on and no motion detected, a loud, pulsing tone will play



Logic Flow Chart and Circuit Diagram



Results and Discussion



- The motion detector senses far away, large motion
 - Any pets around would trigger it
- Alarm pulsing is more noticeable than a steady tone
- Button remains pressed down when stove is off
 - Looser knob may not keep button pressed



Future Work



- Sturdier and more precise attachment mechanisms for knob
- More noticeable alarm
- More accessible and flexible installation methods
 - Tether spool for wiring of different lengths
 - Wireless components for flexible placement



References



[1] J. T. A.; "Fire risk and safety for people living with dementia at home: A narrative review of international literature and case study of fire and rescue services in England," Dementia (London, England), <https://pubmed.ncbi.nlm.nih.gov/39947631/> (accessed Mar. 9, 2026).



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